CHAGEE

# CHAGEE SINGAPORE

## FORECAST THE PEAK, SHARE THE COUNTER

PREDICTIVE AND PHYSICAL AI: WHERE TO OPEN NEXT, AND WHAT THE GM DECIDES FOR THE BUSIEST HOUR

OPIM639 · GROUP 3 · KEZIAH, JIN, AYUMI, REVAN · OCTOBER 2026



# Agenda


01 **Chagee in Singapore**
How it earns and sells

02 **From farm to cup**
Where the supply chain is stable, and where it is not

03 **The data trail**
What every cup already records

04 **Our scope**
Two areas, two kinds of AI, one decision-maker

05 **Predict demand**
App searches, DBSCAN and a new-store pin

06 **Share the counter**
The bucket brigade, side by side

07 **Split the counter**
Which steps stay human: the GM's call

08 **Machine ladder**
What machines exist, what they cost, when to add them

09 **The GM's playbook**
Triggers, pilot and limits

10 **The size of the benefit**
Orders kept, revenue, and what it rests on




# In Singapore, outlets run under one company, and every channel ends at one counter

*One company and one data system: the counter, the app data and the call on who works the peak sit together (informal source, to confirm).*

HOW A SINGAPORE CUP IS SOLD · STEP THROUGH

GROUP NET REVENUE, FY 2025 (SECONDARY SOURCE)




Franchise **88%** Company-owned **12%**, growing fastest

SG Singapore outlets run under one company and one data system (informal source, to confirm with Chagee)

| | |
|---|---|
| **In store** | walk-in, counter or kiosk |
| **Chagee app** | order ahead, collect |
| **Grab** | third-party delivery |

**One counter**
every channel's cup is made here




44 outlets, ground-floor mall units by preference; kiosks in basements and MRT stations at entry in 2019

- **One operator, one data system**
  Singapore outlets run under one company and one data system, so a counter design or roster rule could be rolled out across outlets.
  A Chagee contact (informal), to confirm with Chagee
- **App-first**
  Most orders come through the app, with few kiosks in store. Every order and search is logged: the raw material for a forecast.
  Team observation, to confirm
- **Why Chagee**
  A fast-growing, playful brand with heavy app use: plenty of data, and room for AI to help the stores.

**TAKEAWAY** | Whatever the channel, every order ends at one counter. The peak hour is where an outlet wins or loses.

# Supply is standard and planned ahead; the counter feels demand within minutes

*Tea leaves travel one standard route. Only fresh milk and fruit are bought locally.*

FOLLOW ONE CUP · STEP THROUGH



### Source

Tea leaves from Yunnan through standard channels for every market

DATA CREATED

Shipments, lead times

Local: fresh milk, fruit

### Warehouse

Separate central and regional warehouses keep storage and logistics costs down

DATA CREATED

Stock vs par, expiry

### Outlet counter

Machines brew each cup from a scanned recipe; people do ice, toppings and hand-over

DATA CREATED

Cups by hour

Wait times

### Last mile

Self-collect from the app, or delivery via third-party apps such as Grab; riders wait at the same counter

DATA CREATED

Orders by channel

Rider wait

### Customer

Orders in the app, collects at the counter, comes back (or not)

DATA CREATED

App orders, drop-offs

- **Stable upstream**
  Standard sourcing, regional warehouses and forecast-led restocking: Chagee already plans this part.
  YF Chuhai; Chagee Form 20-F
- **Volatile downstream**
  Demand at a counter moves by the hour: lunch, a mall event, rain, a new drink launch.
- **So we stop at the counter**
  The open question is not the shipment. It is how many people the counter needs this hour, and how they work.

**TAKEAWAY** | Chagee already plans its stock. The open problem is matching people to the hourly peak.

# Every cup is already a timestamped record: the inputs for a forecast exist today

*Four data sources and six decisions they could feed. We follow the four that start in the app and end at the counter.*

WHICH SOURCE FEEDS WHICH DECISION · STEP THROUGH

| | Store and production | App and members | Delivery | Supply and context |
|---|---|---|---|---|
| **Demand forecasts** | Cups per hour by drink and channel | Member repeat demand | Orders per hour by platform | Weather, holidays, launches |
| **Staffing and prep** | Wait times, machine logs | Pickup no-shows | Hand-over peaks | Labour hours vs cups per hour |
| Replenishment | Drink mix per outlet | Launch trial rates | Group-order toppings | Ingredient use, expiry, par |
| **Queue and capacity** | Wait time, footfall vs sales | Store-select drop-offs | Rider wait at counter | Stock-outs at peak |
| **Site selection** | Sales density per outlet | "No store nearby", home and work | Distance and fee by postcode | Mall footfall, competitors |
| Menu and promos | Drink share, customisation | Trial and repeat, voucher uplift | Delivery basket mix | Collab calendar |

White chips mark the core input; grey text refines it. Red rows: our scope.

- **No new sensors needed**
  Point-of-sale, app and machine logs already exist. The work is joining them by outlet and hour.
- **The app is the link**
  Only the app ties a cup to a person. That needs consent and care under Singapore's data protection law (PDPA).
- **What we take forward**
  Demand forecast, then staffing, then queue capacity: one chain that ends with the GM's counter and roster.

**TAKEAWAY** | The data for an hourly forecast is already collected. What is missing is turning it into a roster.

04 · OUR SCOPE

# New stores and the in-store peak, for one decision-maker: the GM

*Two of our three ideation areas, enabled by two kinds of AI, each simulated with the trade-off the GM weighs.*

FROM THREE IDEAS TO TWO · STEP THROUGH

FUTURE WORK

### Online retailing

Steer orders between app pickup and Grab by hour.

Needs platform fees and price tests we do not have.

OUR FOCUS · EXPANSION

### New stores

Map where the next outlet adds the most cups, from app searches.

Rent, fit-out and payback stay future work.

Topic 3 · forecasting

OUR FOCUS · OPERATIONS

### In-store operations

Forecast the peak, set the roster, share the counter.

Data exists, the decision is daily.

Topic 3 · forecasting

Topic 5 · bucket brigades

AI · PREDICTIVE

Forecast demand: where a new outlet adds cups, and cups per hour for the roster.

AI · PHYSICAL

Automation and machines at the counter, from dispensers up to machines that make the drink.

OPERATIONS TOOLS, NOT AI

Bucket brigade and cross-training: how people share the hand steps.

### The GM

General manager, Chagee Singapore

| | |
|---|---|
| DECIDES | Where to open next, which counter steps go to machines, how many people work the peak, and how they share it |
| WHEN | Before a store opens, and when rosters are set |
| USES | The forecasts and the simulations |
| TODAY | To confirm with Chagee: how rosters are set now |

SIMULATED, WITH THE TRADE-OFF THE GM WEIGHS

| | |
|---|---|
| PREDICT | cups added vs cups moved from nearby outlets |
| SHARE | bucket brigade vs fixed posts |
| SPLIT · MACHINES | shorter wait vs higher cost |
| BENEFIT | orders kept vs extra labour |

**TAKEAWAY** | Predictive AI says where to open and how many people the peak needs; physical AI changes which steps machines do; the bucket brigade decides who does the rest.

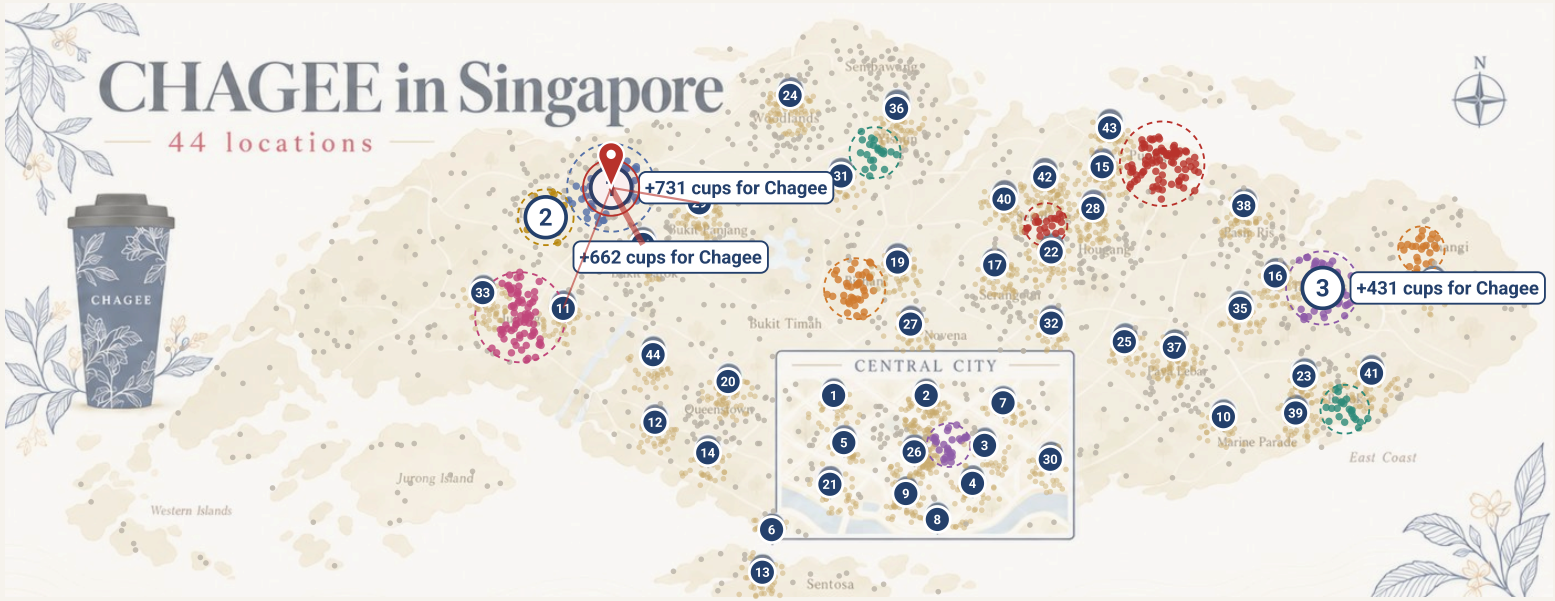
Course link · Week 3, Analytics and AI · the Flashion new-store forecast

# Where would outlet 45 add the most cups for Chagee?

*Not what a new store sells, but what it adds once you subtract the cups it takes from nearby outlets.*

**CONTEXT** Chagee is looking to expand rapidly in Singapore; how many stores it is going for is confidential.

NEW-STORE FORECAST




**SUGGESTED SITE 1 · 2.7 KM FROM 34 · WEST MALL**

**Strong site: mostly new cups**

**+731** cups a day for Chagee

The new store sells 837 a day (1.5× an average outlet). 106 of those move from outlets nearby:

| | |
|---|---|
| 34 · West Mall, 2.7 km | **−77** |
| 29 · Bukit Panjang Plaza, 3.3 km | **−28** |

Peak hour 13:00: **92 cups** → **2 people** on today's counter (slides 06–07).

Synthetic app data on the team's map; outlet positions approximate (the central inset is enlarged). Not a trained model: the point is the method.

**WITHOUT AI** our assumed baseline

Open outlet 45 beside 34 · West Mall: it sells **604** cups a day, but 64% are moved from outlets nearby, so **+216** for Chagee.

**WITH AI** our proposal

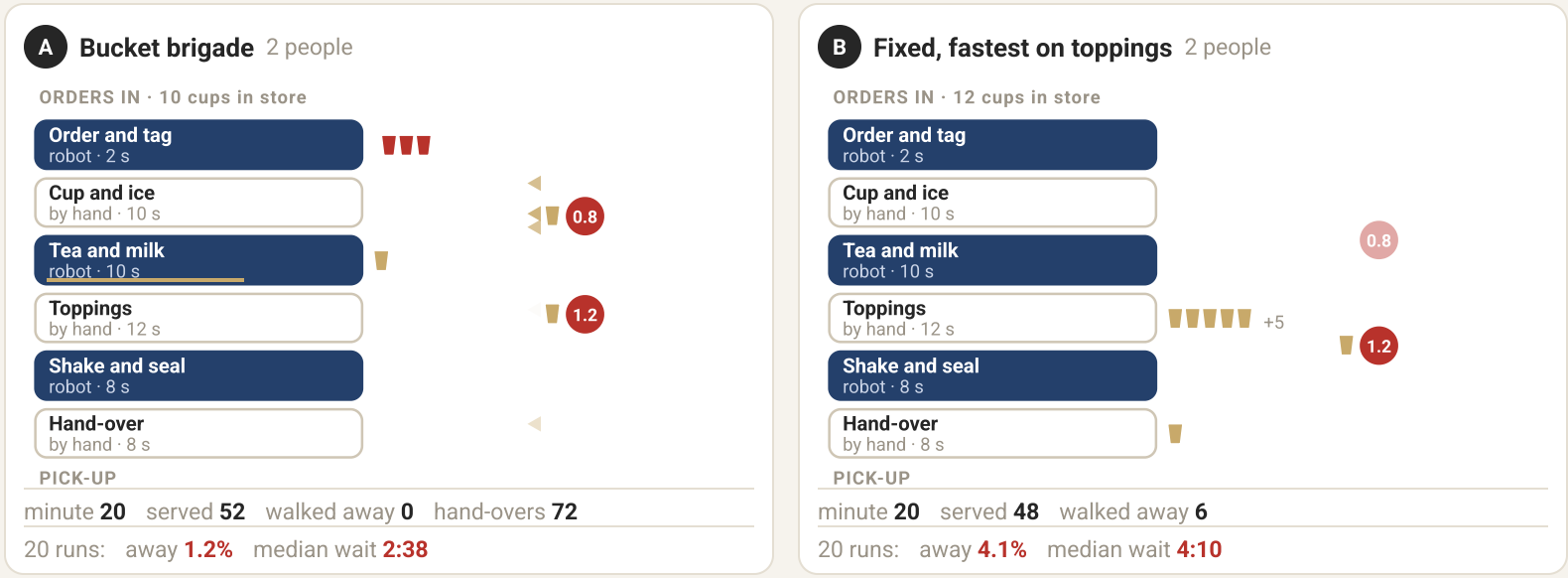
Predictive AI on app searches (DBSCAN) puts the same outlet 2.7 km away, near Choa Chu Kang: it sells 837, **+731** for Chagee.

Course link · Week 5 and Week 6, equipment and technology · the bucket-brigade simulation

# With a lean roster, a bucket brigade keeps cups moving without knowing who is fastest

*Today's counter, two ways of sharing the hand steps, side by side. Pick what each side runs.*

PEAK HOUR · SAME CUSTOMERS ON BOTH SIDES




Red circle = a person (number = speed) · gold ring = held back · faded = walking · ▲ = hand-over · navy = robot · red cup = waiting for a person

ALL THREE RULES, MEAN OF 20 RUNS

| EVENT DAY, 2 PEOPLE | WALKED AWAY | MEDIAN WAIT |
|---|---|---|
| **Bucket brigade** | **1.2%** | **2:38** |
| Fixed, fastest on toppings | 4.1% | 4:10 |
| Fixed, slowest on toppings | 34.9% | 9:47 |

At the people on side A. A customer leaves if the wait would pass 5 min. With spare people, a well-assigned fixed line is as good or a little faster.

- **Why lean is where it counts**
  Work flows to whoever is free, so the line balances itself (Topic 5). Fixed stations are only as good as the guess of who is fast.

**WITHOUT AI** our assumed baseline
A normal-day roster (1 at the peak) on an event day: **54%** of peak-hour orders walk away.

**WITH AI** our proposal
The forecast sets the crew (2) and a bucket brigade shares it: **1.2%** walk away, without knowing who is fastest.

Course link · Week 5 and Week 6, equipment and technology · the bucket-brigade simulation, with machines

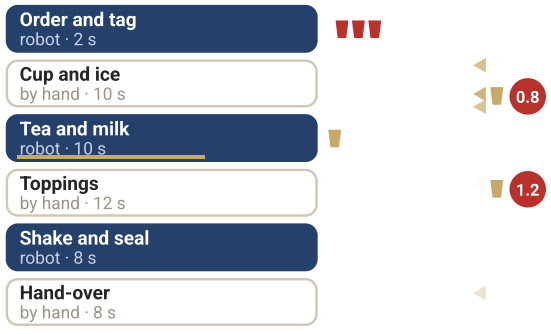
# Which steps stay human? Robots buy a shorter wait; on a normal day, people are still cheaper

*Choose a split for each side, or click any station to hand it to a robot or a person, then run the peak hour.*

PEAK HOUR · SAME CUSTOMERS ON BOTH SIDES

**A Today's counter** 2 people, brigade

ORDERS IN · 10 cups in store

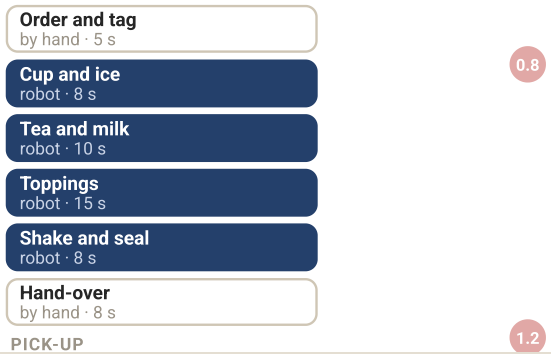



minute **20** served **52** walked away **0** hand-overs **72**

20 runs: away **1.2%** wait **2:38** **S$402**/day

**B Machine-made drink (scenario)** 2 people, fixed posts

ORDERS IN · 0 cups in store




minute **20** served **57** walked away **0**

20 runs: away **0.0%** wait **1:06** **S$465**/day

Navy = robot, white = by hand (click to switch) · red circle = a person · faded = walking · red cup = waiting for a person.

ALL FOUR SPLITS, MEAN OF 20 RUNS

| EVENT DAY | PEOPLE | WALKED AWAY | WAIT | COST/DAY |
|---|---|---|---|---|
| **Today's counter ✓** | **2** | **1.2%** | **2:38** | **S$402** |
| Machine-made drink (scenario) | 2 fixed | 0.0% | 1:06 | S$465 |
| All by hand | 4 | 0.0% | 1:25 | S$422 |
| All robots | – | 0.0% | 0:58 | S$562 |

✓ cheapest within the limit · fastest: All robots

Each split gets the fewest people, from the forecast roster up, and the better sharing rule that keep walk-aways within 2%. Cost per day: people hours at the wage plus robots at their monthly cost.

**Normal day, cost a day:** All by hand S$361 (cheapest) · Today's counter S$372 · Machine-made drink (scenario) S$435 · All robots S$562. Fastest: All robots (0:52); today's counter is slower than all by hand (2:09 vs 1:16). Four splits only; a single machine can undercut all by hand (slide 08).

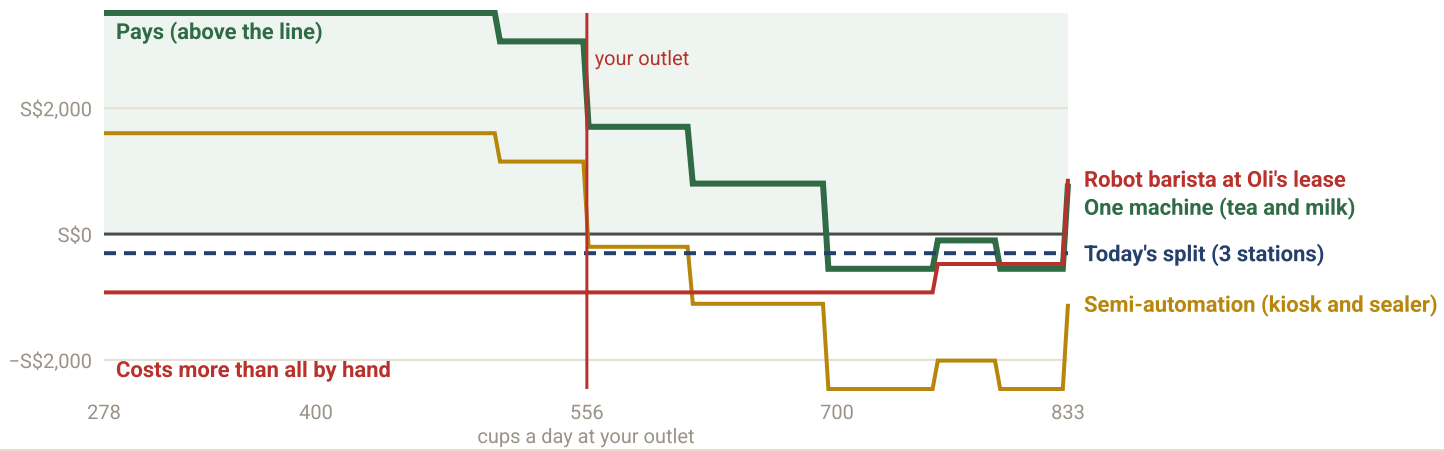
**WITHOUT AI** our assumed baseline — All by hand on an event day: **4 people**, median wait 1:24, **S$422** a day.

**WITH AI** our proposal — Machines on the fixed steps (Today's counter): **2 people**, **S$402** a day, but a longer wait (2:38); all robots wait 0:58 for S$562.

# A machine pays only when it removes a person-shift for less than the shift costs

*To our knowledge, based on public sources; vendor list prices, not cost to implement. Chagee's equipment cost is not public; its Singapore relaunch used semi-automated equipment (Vulcan Post).*

NET GAIN A MONTH AGAINST ALL BY HAND · ONE OUTLET




Shown at 556 cups a day, wage S$15.06 an hour, S$1,909 a month per machine unit (sliders on screen).

| | 1 Semi-automation | 2 Beverage platform | 3 Milk-tea machine | 4 Robot arm | 5 Robot barista |
|---|---|---|---|---|---|
| | Gong Cha Singapore | Gong cha Super Wu | Amacha, one shop | Richtech ADAM | Oli; coffee-first |
| | *Not disclosed* | *Not disclosed* | S$10,000 to buy | *Not disclosed, rental only* | US$4,990/mo lease |
| | PRESS | COMPANY CLAIM | PRESS | VENDOR PAGE | VENDOR PAGE |

The model's S$1,909 a month is per station unit (our reading: a semi-automated station). Semi-automation and the beverage platform are priced at it because their real prices are not disclosed; robot prices are per setup. Full table and sources in the appendix.

START HERE

### One machine on the tea and milk

**+S$1,706** a month vs all by hand

Saves 8 person-hours a day (S$3,614 a month) for one S$1,909 unit. Simple payback, if it lets the roster drop one person-shift: 2.8 months on Amacha's S$10,000.

WATCH

### Today's 3-station split

**−S$305** a month vs all by hand

Saves 12 person-hours a day (S$5,422) but pays for 3 units (S$5,726).

WAIT FOR VOLUME

### Robots: arm or barista

**757+** cups a day before they beat today's split on labour

At 556 cups, Oli's lease (S$6,350) is S$928 a month worse; a robot setup would need to cost under S$5,422 a month. The arm's price is not disclosed.

**Model assumption that drives all three:** minimum crew 2 by hand, 1 with machines, off-peak. Station speeds are the model's, not vendor claims. Rosters step in whole people, so results flip with volume and wage.

**WITHOUT AI** — our assumed baseline

All by hand at 556 cups a day: **24 person-hours** a day, **S$10,843** a month of labour.

**WITH AI** — our proposal

One tea-and-milk machine: **16 person-hours** a day, **+S$1,706** a month after its unit cost. Robots wait for about 757 cups a day.

# Choose the split, roster to the forecast, and share the work to suit the split

*Three moves for the GM, each with a trigger, then a pilot to test them.*

WHAT THE FORECAST CHANGES, PEAK HOUR, TODAY'S COUNTER

| DAY | FORECAST PEAK, CUPS PER HOUR | PEOPLE AT THE PEAK | WALK-AWAYS, FORECAST ROSTER AND BRIGADE | WALK-AWAYS, NORMAL-DAY ROSTER |
|---|---|---|---|---|
| **Normal day** | 61 | **1** | 0.9% | 0.9% |
| **Busy day** | 92 | **2** | 0.0% | **20.8%** |
| **Event day** | 153 | **2** | 1.2% | **53.8%** |

**1 Choose the split**

Robots where speed matters more than cost; people where the work varies: tagging and hand-over.

**2 Roster to the forecast**

On today's counter, one more person for about every 83 forecast cups in the peak hour.

**3 Share to suit the split**

A brigade when hand steps sit along the line; fixed posts when machines make the drink.

- **Next step: pilot in 3 outlets**

| Shadow | Roster | Roll out |
|---|---|---|
| forecast only | staff to it | if KPIs hold |

KPIs: cups per labour hour, wait time, walk-aways at the peak. Compare forecast and actual cups weekly.

- **Limits we know**

All figures are illustrative assumptions, not Chagee data. Robot speeds and costs are assumed; breakdowns and breaks are not modelled.

- **To confirm with Chagee**

How rosters are set today, robot costs per station, and whether outlets log wait times and walk-aways.

**WITHOUT AI** our assumed baseline — One roster for every day: **21%** of peak-hour orders walk away on a busy day, **54%** on an event day.

**WITH AI** our proposal — Roster to the forecast (2 on busy, 2 on event days): **0.0%** and **1.2%** walk away.

# Per outlet, about 61 orders kept on an event day and 5 on a busy day; on an illustrative calendar, S$7,539 of revenue a year

*Modelled avoided costs on stated assumptions, not demonstrated savings. Peak hour only; conservative. Prices to our knowledge, based on public sources.*

Profit a year, 44 outlets: **S$1.09m** = counter automation, physical AI **S$1.04m** (depends on the minimum-crew rule) + forecast roster, predictive AI **S$0.05m** | Without / with the minimum-crew rule: **S$0.37m to S$1.09m**

## ONE OUTLET AT 556 CUPS A DAY · BOTH LEVERS, ONE RUN

**EVENT DAY**
**61** orders kept
Revenue S$358 (× S$5.90) · profit S$241
Counter: labour saved − machine S$27

**BUSY DAY**
**5.4** orders kept
Revenue S$32 (× S$5.90) · profit S$76
Counter: labour saved − machine S$57

**NORMAL DAY**
**0.2** orders kept
Revenue S$1 (× S$5.90) · profit S$58
Counter: labour saved − machine S$57

| A YEAR | PER OUTLET | 44 OUTLETS, IF ALL LOOKED LIKE THIS MODEL |
|---|---|---|
| **Orders kept** peak hour, busy and event days | **1,278** range −121 to 1,501 | **56,221** |
| **Revenue** orders × price | **S$7,539** range −S$832 to S$10,357 | **S$331,704** |
| Profit margin 50%–70%: −S$754 to +S$754 | S$24,660 | S$1,085,027 |
| Labour saving, counter labour saved − machine cost; shown on its own | S$20,136 | S$886,005 |

Revenue and labour are never added: revenue is orders × price; profit is orders × price × margin, plus labour saved, minus machine cost.

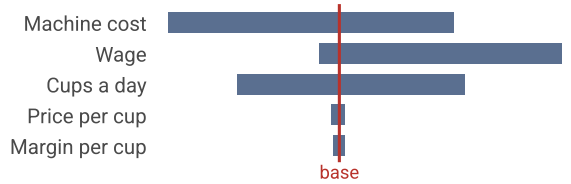
Calendar: 104 busy days (weekend days) and 11 event days (Singapore public holidays) a year: an illustrative definition, to confirm with Chagee; sliders on screen.

| ORDERS · PROFIT A YEAR | WALK-AWAY RULE | ECONOMIC RULE |
|---|---|---|
| Forecast crew | 1,242 · S$603 | 670 · S$1,711 |
| Machine | 888 · S$23,611 | 888 · S$23,611 |
| **Both, one run** | **1,278 · S$24,660** | **1,278 · S$24,660** |
| Both, today's split counter −S$305/month | 1,124 · −S$3,142 | 1,124 · −S$3,142 |

Walk-away rule: fewest people keeping peak-hour walk-aways within 2% (the forecast crew at 556 cups a day). Economic rule: add a person only when orders kept × price × margin beat the extra labour. "Both" is one run, not a sum: the levers keep many of the same orders, and with the machine the forecast crew needs fewer extra hours (+S$446 against the sum).

**Against what**
A flat normal-day roster and an all-by-hand counter. We do not know how Chagee sets rosters today (to confirm with Chagee).

**All inputs at their worst/best end together**
Profit a year, one outlet: low **−S$15,704** · base **S$24,660** · high **S$101,166**




**Fragile: the minimum crew**
The counter lever rests on the model's minimum crew off-peak: 2 by hand, 1 with machines. Profit a year per outlet: S$24,660 with it; S$8,395 with 1 either way.

**Not sized**
Delivery levers (third-party; channel margins not disclosed), membership and CRM, waste (no inventory model in this deck), and the new-store site forecast (shown as cups added only).

**WITHOUT AI** our assumed baseline
Flat normal-day roster, all by hand: about **1,291** peak-hour orders a year walk away per outlet.

**WITH AI** our proposal
Both levers, one run: **1,278** kept a year per outlet, profit **S$1.09m** across 44 outlets (S$0.37m without the minimum-crew rule).

# THANK YOU

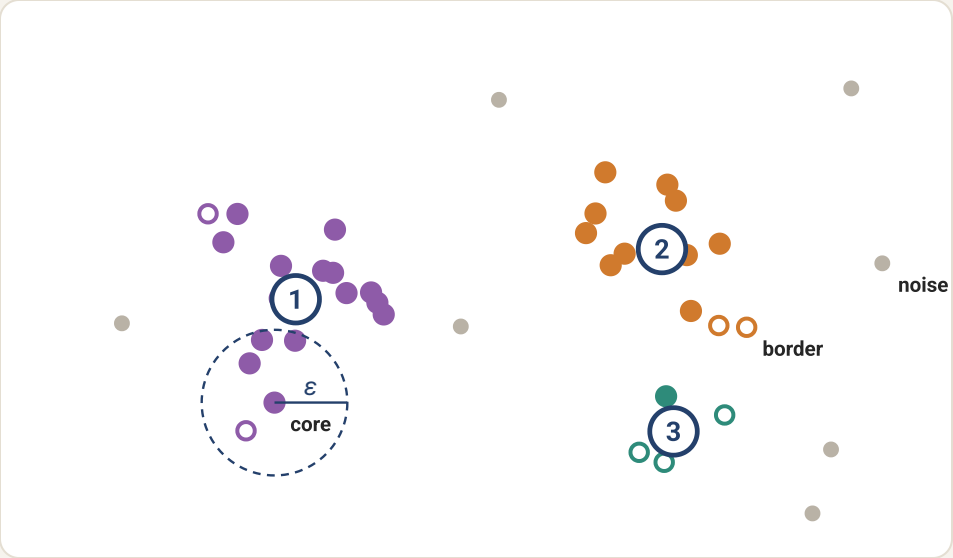
## QUESTIONS AND DISCUSSION

OPIM639 · GROUP 3 · KEZIAH, JIN, AYUMI, REVAN



# DBSCAN finds where searches bunch up; a proximity model turns a pin into cups

## DBSCAN IN FOUR MOVES · TOY EXAMPLE




1. **Radius ε.** Two searches are neighbours if they are within ε of each other (the cluster-radius slider).
2. **Core.** A search with at least the minimum number of searches within ε (counting itself) is a core point.
3. **Grow.** A cluster spreads from core to core. Searches it reaches that are not core join as border points.
4. **Noise.** Searches no cluster reaches. Unlike k-means, nobody picks the number of clusters, and odd shapes are fine.

Unsupervised: no labels or past openings are needed. Ester, Kriegel, Sander and Xu (1996). On slide 05 it runs only on searches with no outlet within walking distance; each cluster's centre is a candidate site.

## THE SIMULATED NUMBERS, IN ORDER · AT THE SLIDER DEFAULTS

| | | | |
|---|---|---|---|
| 1 | **App searches** | Synthetic dots on land: 15% spread evenly, 35% around the outlets' malls, the rest around 20 towns | **3,000 dots** |
| 2 | **No store nearby** | The nearest outlet is further than the walking distance | **1 km · 46% of dots** |
| 3 | **DBSCAN** | Clusters the no-store-nearby dots | **ε 0.6 km · min 15 · 11 clusters** |
| 4 | **Pull of a store** | Fades with distance d: exp(−½ (d/L)²), with the length L set by the walking distance | **L = 1 km** |
| 5 | **Who buys where** | A dot buys at a store in proportion to its pull, against the pull of not buying (a store next door = one) | **not buying 0.3** |
| 6 | **Calibration** | Each dot's weight is set so the average outlet sells the deck's cups a day (cups a year ÷ open days) | **556 cups a day** |
| 7 | **Score a pin** | Cups for Chagee = what the new store sells − what it moves from nearby outlets | **top: 837 − 106 = +731** |
| 8 | **Verdict** | Share of the new store's cups that are new to Chagee | **strong ≥ 60% · weak < 40%** |
| 9 | **Opening roster** | Peak-hour cups ÷ cups one person handles, with the safety margin | **one per 83 cups** |

All synthetic: the dots, towns and weights are assumptions on the team's map (not to scale), not Chagee app logs or population data. With real app logs, steps one and two change; the rest runs unchanged.

APPENDIX · MACHINE LADDER, TIERS AND SOURCES (SLIDE 08)

# Every rung, every price as published, and where it comes from

*To our knowledge, based on public sources. Physical AI: automation and machines. A gap is "not disclosed", never a guess.*

| RUNG | EXAMPLE | AUTOMATES · STEPS ON MACHINES (OUR MAPPING) | VENDOR LIST PRICE | CLAIMS (NOT MODEL INPUTS) | RELIABILITY | SOURCE |
|---|---|---|---|---|---|---|
| 1 **Semi-automation** | Gong Cha Singapore | Sugar dispensers, cup sealers, self-order kiosks. **Order and tag, Shake and seal**. Kiosk = order and tag, sealer = shake and seal. Sugar dispensing sits inside tea and milk, so it is not modelled. | *Not disclosed* | Over 20% productivity from self-ordering; 5% manpower saving (no baseline given) | PRESS | https://www.ntuc.org.sg/uportal/news/Technology-Brewing-Up-the-Bubble-Tea-Business/ |
| 2 **Beverage platform** | Gong cha Super Wu (Feb 2026) | Dispenses tea, sweeteners and syrups; brewing and shaking stay manual. **Tea and milk** | *Not disclosed* | Up to 65% higher peak productivity | COMPANY CLAIM | https://marketech-apac.com/gong-cha-launches-gong-cha-2-0-to-modernise-global-operations-and-store-design/ |
| 3 **Single automated milk-tea machine** | Amacha (one small shop, owner interview) | Makes the milk tea itself. **Tea and milk** | S$10,000 to buy | The shop's total opening cost was S$209,000 (not a machine cost) | PRESS | https://www.malaymail.com/news/life/2026/10/05/opening-a-milk-tea-shop-in-singapore-can-rack-up-to-s209000-even-before-the-first-cup-is-poured/237715 |
| – | **Our scenario: today's split** | **Order and tag, Tea and milk, Shake and seal**. A scenario of ours (slides 06 and 07), not Chagee's disclosed setup | Model: S$1,909 a month per station unit | | OUR ASSUMPTION | assumptions.json |
| 4 **Robot arm that makes boba** | Richtech ADAM | Cup, ice, tea, toppings and sealing by one robot arm. **Cup and ice, Tea and milk, Toppings, Shake and seal** | *Not disclosed*. Rental only; price not published | – | VENDOR PAGE | https://richtechrobotics.com/solutions/adam |
| 5 **Robot barista** | Oli | A whole drink station in one unit. **Cup and ice, Tea and milk, Toppings, Shake and seal**. Coffee-first. No tea drinks listed for Oli or Artly. | US$59,000 (≈ S$75,078) to buy; US$4,990 (≈ S$6,350) a month lease | – | VENDOR PAGE | https://olirobotics.com/best-robot-barista |
| | Artly | | US$68,000–88,000 (≈ S$86,530–111,980) to buy; US$2,500–4,000 (≈ S$3,181–5,090) a month lease; US$2,500–4,000 a month software<br>Oli's estimate of a competitor. The figures as given do not say whether the lease replaces or adds to the purchase, so no verdict is computed. | – | VENDOR PAGE | https://olirobotics.com/best-robot-barista |
| | Cafe X | | US$250,000 (≈ S$318,125) to buy<br>Oli's estimate of a competitor's price. | – | VENDOR PAGE | https://olirobotics.com/best-robot-barista |
| – | **Chagee Singapore** | Chagee's own equipment cost is not public. Its Singapore relaunch used semi-automated equipment. | *Not public* | – | PRESS | https://vulcanpost.com/872176/chagee-singapore-comeback-after-sudden-exit/ |

**Vendor list prices,** not cost to implement: installation, training, maintenance, power, space and import are not included; USD converts at S$1.2725 per US$. **Simple payback** = list price ÷ labour saved a month, with no maintenance or financing. **The model's S$1,909 a month** is per station unit: our reading: one semi-automated station unit (for example a dispenser or a cup sealer), amortised with upkeep. Not a robot barista lease; robot prices in machine_tiers.json are per setup. **Vendor speed and throughput claims** are listed, never used: every rung runs at the model's station speeds. **Steps on machines** is our mapping. Artly and Cafe X prices are Oli's estimates of competitors.

# Where every number comes from

## FACTS QUOTED

**Outlets in Singapore: 44**. Store count used on the storyboard deck. The doc also cites 46 (VnExpress, earlier); the later directory count is used. *[L-sg-directory]*

**Franchise share of FY2025 net revenue: 0.88**. Secondary source; flagged wherever shown. *[L-koalagains]*

**Company-owned share of FY2025 net revenue: 0.12**. Secondary source; flagged wherever shown. *[L-koalagains]*

## SOURCES

**L-sg-directory** Chagee Singapore store directory, mapped by the team (44 locations, checked 15 Oct 2026); Mothership, Sep 2026, https://mothership.sg/2026/09/chagee-more-heartland-stores-singapore-eugene-lee/ gives the same count. Taken from the team's ideation doc (OPIM 639_ Ideation Process), not re-checked for this deck.

**L-koalagains** KoalaGains, 'CHA business and moat', https://koalagains.com/stocks/NASDAQ/CHA/business-and-moat (SECONDARY - flag when used): FY2025 net revenue about 88% from the franchise (master franchise) model and about 12% from company-owned stores, the latter growing fastest. Taken from the team's ideation doc (OPIM 639_ Ideation Process), not re-checked for this deck.

**L-yfchuhai** YF Chuhai, https://www.yfchuhai.com/article/9751.html: Chagee's leaf supply chain is in China; it sources only two ingredients locally, fresh milk and fruit. Taken from the team's ideation doc (OPIM 639_ Ideation Process), not re-checked for this deck.

**L-sec20f** Chagee Holdings Form 20-F for FY2025, SEC EDGAR, https://www.sec.gov/Archives/edgar/data/2013649/000110465926050766/cha-20251231x20f.htm: separate centralized and regional warehouses to lower storage and logistics costs. Taken from the team's ideation doc (OPIM 639_ Ideation Process), not re-checked for this deck.

**L-unsw** UNSW BusinessThink, Oct 2024: Chagee semi-automation, baristas scan a QR code and the machine dispenses the formulation. (brief.md section 9)

**I-chagee-contact** INFORMAL - a Chagee contact (Oct 2026). Three points only: whether Singapore outlets are owned, a joint venture or franchised does not matter for this work; Singapore outlets run under one company and one data system; all delivery is via third-party apps. Confirm with Chagee before citing.

**I-sg-groundfloor** INFORMAL - quote in the team's ideation doc (Chagee Singapore's Lee: 'Whenever you see a Chagee, it's always on the first floor'; kiosk-style outlets in basements and MRT stations at entry in 2019). The article is not named in the doc; confirm the source before quoting.

**I-team-app-first** INFORMAL - team observation in the ideation doc: Chagee Singapore sells mainly through its app, with minimal order kiosks in outlets. Confirm before citing.

**L-ester1996** Ester, M., Kriegel, H.-P., Sander, J. and Xu, X. (1996). A density-based algorithm for discovering clusters in large spatial databases with noise. Proceedings of the Second International Conference on Knowledge Discovery and Data Mining (KDD-96), 226-231. The DBSCAN algorithm used on slide 05.

**P-honeycombers** PRESS - in-store sightings, indicative. Honeycombers (Feb 2026), https://thehoneycombers.com/singapore/china-bubble-tea-brand-singapore/ : Chagee price sightings in Singapore; not a current full menu.

**P-eatbook** PRESS - in-store sightings, indicative. Eatbook (Aug 2024), https://eatbook.sg/?p=168476 : Chagee price sightings in Singapore; not a current full menu, and older than the deck.

**I-team-expansion** INFORMAL - team research (Oct 2026): Chagee is looking to expand rapidly in Singapore; how many stores it is going for is confidential. Singapore only. Confirm before citing.

## ASSUMPTIONS (ASSUMPTIONS.JSON, ILLUSTRATIVE)

**Cups per outlet per year**: 200000 cups/store/year

**Share of the day's cups in each open hour**: 0.04, 0.06, 0.1, 0.11, 0.08, 0.08, 0.09, 0.1, 0.1, 0.09, 0.08, 0.07 share of daily cups per open hour (10:00 to 22:00), sums to 1

**Busy and event day multipliers**: none 1, moderate 1.5, large 2.5 x forecast arrival rate

**Busiest hours the multiplier lands on**: 2 busiest forecast hours that get the surge

**Seconds per cup by hand, per station**: ticket 5, cup ice 10, tea milk 15, topping finish 12, shake seal 10, handover 8 seconds of one average worker per cup

**Seconds per cup by robot, per station**: ticket 2, cup ice 8, tea milk 10, topping finish 15, shake seal 8, handover 6 seconds per cup per machine unit

**Stations the robots do today**: ticket, tea milk, shake seal

**Spread of worker speeds**: 0.2 +/- share around average speed

**Minute-to-minute variation in pace**: 0.25 coefficient of variation, minute to minute

**Share of new staff**: 0.25 share of roster in first months

**New staff speed**: 0.7 x average speed

**Coordination loss, people around robots**: 0.9 multiplier on human-pool throughput in the mixed setup

**Minimum people on the counter**: all human 2, mixed 1, all machine 0 workers per open hour

**Roster safety margin over the forecast**: 1.2 x forecast peak rate

**Labour cost per hour worked (SGD)**: 15.06 SGD per hour worked, fully loaded (statutory costs)

**Robot cost per station unit per month (SGD)**: 1908.75 SGD per station machine unit per month

**Attendant hours, all-robot counter**: 1 workers per open hour

**Walk-away limit for the GM's split table**: 0.02 share of peak-hour orders that walk away

**Synthetic app-intent dots (slide 05)**: 3000 synthetic app-intent dots on the map

**Walking distance to collect**: 1 km a customer will walk to collect

**Distance decay, share of the walking distance**: 1 distance decay length as a share of walk_km

**Pull of not buying**: 0.3 pull of not buying, against a store next door (= 1)

**DBSCAN cluster radius**: 0.6 km

**DBSCAN minimum searches**: 15 dots within eps to form a cluster core

**Map scale**: 0.033 km per map pixel

**Wait at which a customer walks away**: 5 minutes

**Price per cup, in store (slide 10)**: 5.9 SGD per cup, in store

**Margin per cup, share of price (slide 10)**: 0.6 share of price (contribution per avoided order: price minus ingredients and packaging)

**Busy days a year (slide 10, illustrative)**: 104 busy days a year (day type 'moderate')

**Event days a year (slide 10, illustrative)**: 11 event days a year (day type 'large')

**Walking-back speed (Week 5 units)**: 6 lines per hour

## CLAIMS WE DID NOT MAKE

No Chagee sales, staffing or wait-time data is used; the forecast is not a trained model. We do not claim the bucket brigade always beats fixed stations: with spare staff, a well-assigned fixed line is as good or slightly faster. Machine breakdowns, breaks and order mix are not modelled.

We make no claim about: **Singapore ownership structure** (a Chagee contact says outlets run under one company and one data system, and that the ownership form does not matter here; I-chagee-contact); **delivery profit by channel**; **Chagee's own equipment cost** (not public); **gross margin per cup** (our assumption); **Chagee's current rostering** (unknown; slide 10's counterfactual is ours).